Contrasting Estimation of Pattern Prototypes for Anomaly Detection in Urban Crowd Flow

Yupeng Wang, Xiling Luo, and Zequan Zhou

***Abstract*— Crowd flow anomaly detection (CFAD) plays a vital role in ensuring public safety due to its capability to distinguish abnormal crowd movement behaviors from the norm. However, the influence of coexisting spatiotemporal factors presents a substantial challenge in capturing the dynamic normal pattern. Moreover, the inherent similarities within the original crowd flow data necessitate the creation of a discriminative feature space. To address this, we propose *ProtoDetect*, a novel method that learns distinguishable representations (named as prototypes) of the influencing factors. It subsequently identifies anomalous samples by comparing them with their normal counterparts, estimated based on the prototypes. Experimental evaluations on three real-world datasets demonstrate ProtoDetect's consistent superior performance in CFAD. The source code is available at https://github.com/yupwang/ProtoDetect.**

***Index Terms*— Anomaly detection, crowd management, urban transportation systems, contrastive learning.**

I. Introduction

CROWD management plays a crucial role in enhancing urban transportation systems, traveler information systems, and promoting sustainable urban development [1], [2], [3], [4]. With the advancement of urban transportation infrastructures, large volumes of traffic data is being collected from sources such as vehicle trajectories, trip records, mobile phone call logs, social media posts, and surveillance camera signals [4], [5], [6]. This data provides valuable insights into individual movement within urban environments and has expedited the development of efficient crowd management in areas such as crowd simulation [1], [7], [8] and crowd flow prediction [4], [9], [10]. Nonetheless, further investigation is required to address the field of anomaly detection within crowd flow.

Crowd flow anomaly detection (CFAD) involves identifying uncommon or abnormal patterns in the movement of individuals within urban environments. Crowd flow refers to the count of people entering (inflow) or exiting (outflow) specific urban regions during specific times, as shown in Figure 1. Anomalies in crowd flow arise from exceptional events. For example, occurrences such as sports competitions, festivals, and concerts often result in increased crowd flow. Conversely, extreme weather conditions typically lead to a decrease in crowd flow. Street closures, on the other hand, result in a decrease in the crowd flow of the closed street and an increase in neighboring streets [11]. Recognizing anomalies in crowd flow is essential as it helps identify these issues and is crucial for enhancing public safety and improving urban planning.

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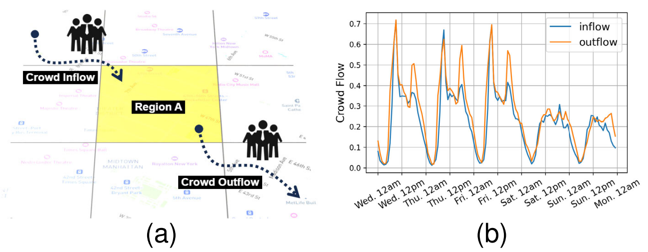


Fig. 1. (a) Illustration of inflow and outflow of Region A. (b) Crowd flow variation from Wednesday to Sunday (eight-week average).

Performing CFAD encounters two main difficulties. Firstly, due to the complex impact of anomalies on crowd flow, simply recognizing decreases or increases in the original crowd inflow and outflow struggles in real-world environments. To illustrate this challenge, we examine the inflow and outflow of an urban region encompassing Millennium Park and Grant Park in Chicago during the period from August 16th, 2019, to September 1st, 2019, as depicted in Figure 2. We estimate crowd flow dynamics by aggregating taxi and bike trip records arriving at or leaving the region every hour (as detailed in Section V-A). Real-world anomalies happened in the two parks, including Grant Park Music Festival (16th, 17th), Summer Dance Celebration (24th), and Chicago Jazz Festival (29th, 30th), are indicated by light red shading. The observation from the figure reveals the considerable challenge in discerning a distinct and intuitive difference in crowd flow patterns in the presence or absence of anomalies.

Secondly, modeling the normal patterns that represent the expected crowd behavior poses a difficulty. In anomaly detection, it is common to compute the deviation of observed data from their expected normal pattern. However, estimating the normal pattern is complex as it involves contributions from various spatial and temporal factors. Temporal factors include considerations like the time of day and day of the week. For example, crowd flow tends to peak during morning and evening hours while significantly decreasing during nighttime. Similarly, crowd flow in office buildings may be much more active on workdays and quieter on weekends. Holidays introduce additional temporal factors, leading to

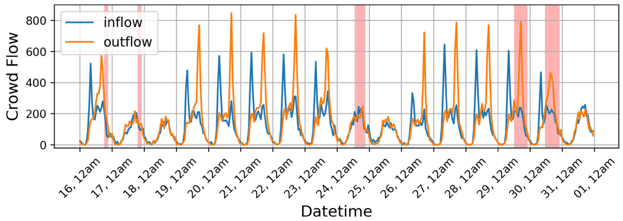


Fig. 2. Crowd flow of the region encompassing Millennium Park and Grant Park, Chicago, from August 16th to September 1st, 2019. Light red shading highlights real-world anomalies in the two parks, including the Grant Park Music Festival on the 16th and 17th, the Summer Dance Celebration on the 24th, and the Chicago Jazz Festival on the 29th and 30th, emphasizing the challenge in distinguishing crowd flow patterns in the presence of anomalies.

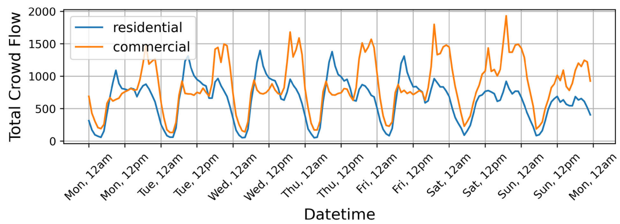


Fig. 3. Eight-week averaged total crowd flow (sum of inflow and outflow) of regions with different functional properties. Various factors contribute distinctively to the normal pattern.

increased crowd flow in tourist areas and decreased flow in work-related regions. Spatial factors, such as the presence of public functional places, also contribute to diverse crowd flow patterns. To illustrate this, we present the eight-week averaged total crowd flow (sum of inflow and outflow) of two urban regions in Figure 3. These regions are categorized as residential and commercial based on the number of functional places within them. It is evident that the peak hours of the two regions exhibit noticeable differences. Additionally, the crowd flow in the residential area experiences a substantial decrease on weekends, whereas the flow in the commercial region is more evenly distributed throughout the week.

Existing approaches for CFAD, including general traffic anomaly detection methods, typically adopt a two-step approach. Firstly, they transform the original traffic data into a feature space. These features encompass direct traffic dynamics such as vehicle speed, travel distance, and traffic flow volume [12], [13], as well as handcrafted statistics derived from the dynamics [14], [15], [16]. Advanced techniques like tensor factorization [17] or neural networks [11], [18] can also be employed for feature extraction. In the second step, general anomaly detection algorithms widely used in data science are applied to the extracted features to identify anomalous data points. However, despite the effectiveness of these approaches, they fail to adequately address the two difficulties mentioned previously. Here we summarize the aspects that have not been adequately considered:

1) *Homogeneity of the feature space:* As discussed in Figure 2, the original features (inflow and outflow) tend to exhibit similarities, making it challenging to identify anomalies. Existing approaches address this issue by utilizing handcrafted statistics derived from the original features [14], [15], [16], [19], or by employing embedding features obtained through tensor factorization [17] or neural networks [11], [18]. However, these feature generating processes often overlook the spatiotemporal dependencies of data points and do not prioritize enhancing feature discriminability. The homogeneity among features makes it difficult to effectively distinguish anomalies from normal data points.
2) *Coexistence of spatiotemporal factors:* Capturing the expected crowd behavior necessitates modeling the spatial and temporal factors and how they contribute to normal patterns. However, existing approaches often rely on general anomaly detection algorithms that are not specifically designed to handle the simultaneous occurrences of influencing factors. This oversight leads to inaccurate estimations of normal patterns and can hinder the detection of anomalies.

In this paper, we leverage contrastive representation learning to obtain a discriminative feature space, and to represent distinguishable influencing factors. Contrastive representation learning has garnered significant attention in recent years, primarily due to its outstanding performance across diverse domains, including computer vision [20], [21], [22], time series analysis [23], [24], [25], and graph learning [26], [27]. This approach, executed in an unsupervised manner, aims to foster a representation that effectively distinguishes one instance from others. Specifically, to overcome the homogeneity challenge in the feature space, we introduce the **Instance Contrast Module (IC)** to obtain an informative and discriminative feature space. This module captures spatiotemporal dependencies by contrastively encoding local and global views of crowd flow data. Furthermore, by comparing the encoded samples with their augmented counterparts, it effectively disperses the samples in the feature space, thereby enhancing their discriminability. Additionally, we propose the **Prototype Contribution Contrast Module (PCC)** to estimate representations of the influencing factors in the feature space. These representations are referred to as **prototypes**. Instead of directly modeling the prototypes themselves, we focus on modeling their contribution to the observed crowd flows. To achieve this, we train a projection head that estimates the contributions by contrasting contribution vectors between the local and global views. This process aligns vectors from the same prototype and repels vectors from different prototypes, leading to distinguishable prototype distributions, as demonstrated in our experiment section V-D.3. These contribution vectors are then used to iteratively update the randomly initialized prototypes through a cross-view prediction task. In the anomaly degree scoring stage, we measure the degree of anomaly by computing the deviation between incoming crowd flow features and their corresponding normal features, which are estimated based on the prototypes. Our main contributions can be summarized as follows.

1) We present the ProtoDetect model for CFAD. It introduces the IC module to acquire an informative and discriminative representation of crowd flow data, enabling accurate differentiation between abnormal and normal crowd flow. It introduces the PCC module to facilitate the estimation of prototypes and their contributions to crowd flow, providing a new means of modeling the normal pattern.

2) We conduct extensive experiments on three sets of real-world crowd flow anomalies. The result demonstrates our ProtoDetect method surpasses baseline approaches, showcasing a significant enhancement in precision. Additionally, we conduct case studies to examine the evolution of anomaly scores during the occurrence and conclusion of anomalies. This analysis offers insights into the behavior of the proposed method and its effectiveness in detecting anomalies in real-world traffic environments.

The rest of this article is organized as follows. We first present the related work regarding traffic anomaly detection and contrastive learning in Section II. We then clarify the definitions of key terms used in the paper in Section III. In Section IV we introduce details of our ProtoDetect method. The experiment results are discussed in Section V. Finally, we conclude our work in Section VI.

II. Related Work

In this section, we introduce the existing works in traffic anomaly detection and contrastive learning.

A. Traffic Anomaly Detection

General traffic anomaly detection approaches can be broadly categorized into three types: handcrafted feature-based, tensor-based, and deep learning-based.

Handcrafted feature-based methods involve the creation of features according to predefined rules from traffic data, followed by the application of general anomaly detection algorithms to these features. Commonly employed anomaly detection algorithms in this category include the elliptic envelope (EE) [28], isolation forest (IF) [29], local outlier factor (LOF) [30], and One-Class SVM (OCSVM) [31]. These features may have physical interpretations, such as vehicle speed, travel distances, and movement direction [12], [13], [32]. They can also be manually defined statistics, encompassing aspects like spatial similarity, temporal similarity, and traffic flow distribution [14], [15], [16]. While these features provide valuable insights into real-world implications and are easily interpretable, they often lack high-dimensional spatiotemporal characteristics and rely heavily on, and are limited by, human knowledge.

Tensor-based methods are employed to learn normal patterns in traffic data by utilizing restricted tensor factorization. In this approach, traffic data is represented as a region-feature-time tensor [17], [33], [34], [35]. For instance, [17] utilize nonnegative CP decomposition to estimate normal patterns and their distribution across temporal and spatial dimensions. To identify anomalous regions that display distributions distinct from their past behavior, the LOF algorithm is subsequently employed. Similarly, [33] employ nonnegative tensor cofactorization, incorporating the social activity check-in tensor. However, it is important to note that tensor-based methods model regions independently, neglecting spatial correlations between them.

Deep learning-based methods have demonstrated their efficacy in modeling traffic data [36], [37], [38] and are also used in traffic anomaly detection [11], [18], [39]. Neural networks can operate in two key modes: first, by generating spatiotemporal features that are subsequently fed into general anomaly detection algorithms, as exemplified by ST_decomp [18], which employs fully connected layers to capture spatial and temporal patterns and feeds the deviation from normal patterns into the LOF algorithm for anomaly detection. Second, neural networks can be tailored for end-to-end anomaly detection, as seen in STGAN [11], which employs a generator to produce synthetic traffic data slices and a discriminator to assess the degree of anomaly. By training a generator to produce synthetic slices, it can be argued that STGAN essentially models a single abnormal pattern. However, although these deep learning-based methods are capable of modeling single normal or abnormal patterns, they often overlook the complexity of traffic data. Different times of the day, regions with varying social properties, and weekdays versus weekends can exhibit different normal patterns.

The connections between our work and existing research, as well as our improvements, can be outlined as follows. Like existing feature-based methods, we propose the IC module to generate high-dimensional crowd flow features. However, we go beyond existing approaches by utilizing instance contrastive learning. This enables us to capture spatiotemporal dependencies and obtain a discriminative feature space that improves the detection of anomalies. In our PCC module, we adopt the idea of estimating normal patterns, which is commonly used in tensor-based methods. However, we enhance this concept by estimating prototypes and utilizing the learned prototypes to predict normal components. Additionally, we introduce a scoring method that evaluates the deviation between incoming crowd flow features and their corresponding normal features, which are estimated based on the prototypes. This scoring method is particularly well-suited for the urban traffic environment.

B. Contrastive Learning

In our study, we employ contrastive learning as a technique to acquire discriminative high-dimensional features and prototype contribution vectors. Contrastive learning, a popular approach in deep learning, facilitates unsupervised representation learning by repelling negative pairs while attracting positive pairs [21], [22], [23], [40].

The representations learned through contrastive learning have proven to be effective in various domains such as computer vision (image classification, object detection) [20], time series analysis (human activity recognition, fault diagnosis) [23], and graph learning (node classification, graph classification) [26]. To facilitate effective contrastive learning, data augmentation techniques are commonly employed to create positive and negative pairs. These techniques involve generating different views of the original data, where samples from the same data points are considered positive pairs, while samples from different data points are treated as negative pairs. For image data, common augmentation techniques include cropping, resizing, rotating, or blurring the images [21], [22]. In the case of time series data, transformations in the time and frequency domains are applied [23], [41], [42]. In graph learning, graph diffusion is often utilized to create a global view for contrastive learning [26], [27], [43].

TABLE I
DESCRIPTION OF MATHEMATICAL SYMBOLS

Notation	Description
$\mathcal{V} = \{v_1, v_2, \ldots, v_n\}$	the set of urban regions
$\mathcal{T} = \{t_1, t_2, \ldots, t_l\}$	the set of time slots
γ	timestamp
$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$	urban region network
$\mathbf{A} \in \mathbb{R}^{n\times n}$	adjacency matrix of $\mathcal{G}$
Ω	the set of trip records
$\mathsf{X} \in \mathbb{R}^{l\times n\times 2}$	crowd flow dynamics
$\mathbf{O} \in \mathbb{R}^{l\times n}$	anomaly status matrix
$\mathcal{U}$	the set of anomaly events
$\mathbf{D} \in \mathbb{R}^{n\times n}$	diagonal degree matrix of $\mathbf{A}$
$\mathbf{S} \in \mathbb{R}^{n\times n}$	diffusion matrix
λ	diffusion time
y	spatiotemporal instance
T	length of the time series in y
$f(\cdot)$	function mapping spatiotemporal instances into features
$\mathbf{h} \in \mathbb{R}^{H}$	feature
$\mathcal{L}_{ic}$	instance contrasting loss
τ_{ic}	instance contrasting temperature
N	number of data points in one batch
M	number of prototypes
$\mathbf{Z} \in \mathbb{R}^{N\times M}$	contribution matrix
$\mathbf{c}^{(i)} = \mathbf{Z}_{:,i} \in \mathbb{R}^{N}$	prototype contribution vector
$\mathcal{L}_{pc}$	prototype contribution contrasting loss
τ_{pc}	prototype contribution contrasting temperature
$\mathbf{P} \in \mathbb{R}^{M\times H}$	prototypes
$\mathcal{L}_{pred}$	prediction loss
$\mathsf{X}\text{test} \in \mathbb{R}^{l_{\text{test}}\times n\times 2}$	crowd flow dynamics of the test datset
$\mathbf{W} \in \mathbb{R}^{l_{\text{test}}\times n}$	anomaly score matrix of Xtest
$\mathbf{Q} \in \mathbb{R}^{M\times M}$	similarity matrix of $\mathbf{P}$

In our study, as we represent crowd flow data as a sequence of attributed graphs, we propose a combination of time series augmentation and graph augmentation techniques to create two views of the original data. We construct instances that incorporate spatiotemporal neighboring information from both views and encode them into discriminative feature vectors by contrasting positive and negative instance pairs. Additionally, we model the contribution of prototypes by contrasting contribution vectors from both views.

III. PRELIMINARY

The definitions of key terms used in the paper are shown as follows. Table I lists the mathematical symbols used in the paper.

Definition 1 (Urban Region): Urban regions are denoted by partitioning the bounding box of the urban boundary into a grid of cells, based on longitude and latitude. Each grid cell has uniform dimensions and corresponds to a distinct region. Note that grid cells falling outside the urban boundary are excluded. Formally, we denote the urban region set with symbol $\mathcal{V} = \{v_1, v_2, \ldots, v_n\}$.

Definition 2 (Time Slot): We denote a set of timestamps as $\{\gamma_1, \gamma_2, \ldots, \gamma_{l+1}\}$, with each consecutive pair of timestamps satisfying the condition $\gamma_{i+1} - \gamma_i = 1$hour. The time slot set is formally defined as $\mathcal{T} = \{t \mid t_i = (\gamma_i, \gamma_{i+1}), i \le l\}$, where each time slot represents a 1-hour time interval.

Definition 3 (Urban Region Network): Viewing urban regions as nodes in a graph, we construct the urban region network represented by an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$. This graph describes the interrelationships between urban regions, where $\mathcal{E}$ denotes the set of edges indicating the connectivity between these regions, and $\mathbf{A} \in \mathbb{R}^{n\times n}$ represents the adjacency matrix of $\mathcal{G}$.

Definition 4 (Crowd Flow Dynamics): We model the crowd flow dynamics by aggregating trip records, which detail the initiation and termination times and locations of individual movements. For instance, a trip record can be represented as {(−73.99, 40.73), (−73.98, 40.73), 2014-01-09 20:45, 2014-01-09 21:12}, signifying the starting location, ending location, starting time, and ending time of a specific trip.

We formally define the collection of trip records as $\Omega = \{\omega \mid \omega_i = (v^{\text{trip}}_{s_i}, v^{\text{trip}}_{e_i}, t^{\text{trip}}_{s_i}, t^{\text{trip}}_{e_i})\}$. In this context, $v^{\text{trip}}_s \in \mathcal{V}$ and $v^{\text{trip}}_e \in \mathcal{V}$ represent the starting and ending regions of the trip, respectively. Similarly, $t^{\text{trip}}_s \in \mathcal{T}$ and $t^{\text{trip}}_e \in \mathcal{T}$ represent the time slots during which the trip commences and concludes.

Crowd flow dynamics are represented as a tensor $\mathsf{X} \in \mathbb{R}^{l\times n\times 2}$. The inflow and outflow are defined as:

$$X_{i,j,1} = \left|\left\{\omega \mid \omega_k \in \Omega \wedge t^{\text{trip}}_{e_k} = t_i \wedge v^{\text{trip}}_{e_k} = v_j\right\}\right| \quad (1)$$

$$X_{i,j,2} = \left|\left\{\omega \mid \omega_k \in \Omega \wedge t^{\text{trip}}_{s_k} = t_i \wedge v^{\text{trip}}_{s_k} = v_j\right\}\right| \quad (2)$$

where $X_{i,j,1}$ and $X_{i,j,2}$ denote the inflow and the outflow, respectively.

Definition 5 (Problem: Crowd Flow Anomaly Detection): We represent expert-labeled anomaly events using the symbol $\mathcal{U} = \left\{\left(t_{s_1}, t_{e_1}, v_{r_1}\right), \left(t_{s_2}, t_{e_2}, v_{r_2}\right), \ldots, \left(t_{s_k}, t_{e_k}, v_{r_k}\right)\right\}$. Each individual anomaly event is denoted by a tuple (t_s, t_e, v_r), where $t_s \in \mathcal{T}$ represents the starting time slot of the event, $t_e \in \mathcal{T}$ indicates the ending time slot of the event, and $v_r \in \mathcal{V}$ specifies the region in which the event takes place.

Given crowd flow dynamics denoted as $\mathsf{X} \in \mathbb{R}^{l\times n\times 2}$, our objective is to derive the anomaly status matrix $\mathbf{O} \in \mathbb{R}^{l\times n}$, wherein $O_{i,j} \in \{0, 1\}$ indicates whether a data point at position (i, j) is normal (0) or anomalous (1). The ideal anomaly status matrix $\mathbf{O}^*$ is labeled by anomaly events $\mathcal{U}$, thereby serving as a reference for assessing the accuracy of anomaly detection algorithms, i.e.,

$$O^*_{i,j} = \begin{cases} 1 & \text{if } \exists\,(t_s, t_e, v_r) \in \mathcal{U}, \\ & \text{such that } t_s \le i \le t_e \text{ and } v_r = j; \\ 0 & \text{otherwise.} \end{cases} \quad (3)$$

IV. THE PROPOSED METHOD

The proposed ProtoDetect model consists of two main modules: the IC module and the PCC module, as depicted in Figure 4. The IC module encodes spatiotemporal information of crowd flow in a contrastive manner, enabling it to learn discriminative features from the original data. The PCC module estimates prototypes within the feature space and leverages them to model the normal component of crowd flow. The anomaly degree of incoming data points is determined by measuring their deviation from the estimated normal component, computed using the prototypes.

A. Instance Contrast Module

The IC module utilizes unsupervised contrastive learning to create a discriminative feature space. It achieves this by employing both temporal and spatial augmentation techniques

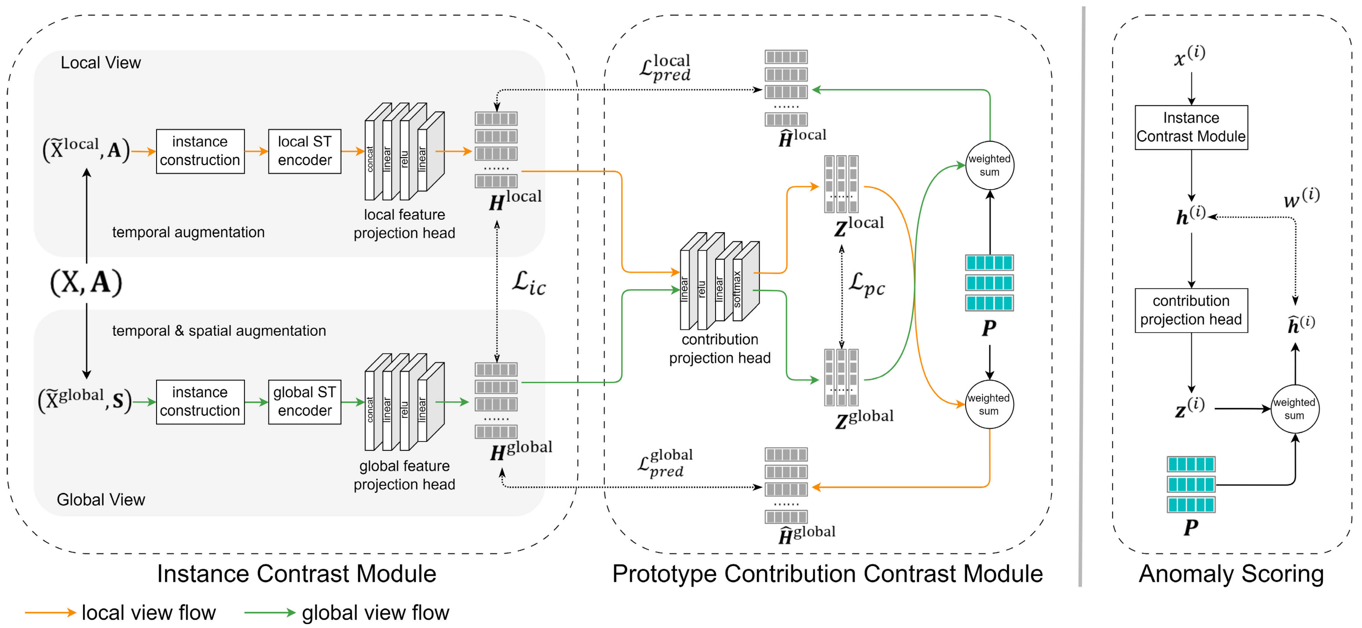


Fig. 4. The proposed ProtoDetect model comprises two modules: the *Instance Contrast Module (IC)* and the *Prototype Contribution Contrast Module (PCC)*. By utilizing spatiotemporal contrastive learning, the IC module extracts discriminative features from the original crowd flow data. Specifically, it generates local and global views by applying temporal and spatial augmentation techniques, respectively. Instances in each view are encoded and projected into features, denoted as $\mathbf{h}$, using a pre-trained spatiotemporal encoder and a feature projection head. The instance contrasting loss $\mathcal{L}_{ic}$ is defined to compare positive feature pairs from the same data point and negative pairs from different data points, thereby enhancing the discriminability of the feature space. Subsequently, in the feature space, the PCC module estimates prototypes $\mathbf{P}$ contributing to the normal crowd flow patterns. Instead of directly modeling the prototypes, it estimates the contribution of prototypes to the observed crowd flow by contrasting contribution vectors $\mathbf{z}$ between the local and global views. This involves training a contribution projection head with the prototype contribution contrasting loss $\mathcal{L}_{pc}$. Moreover, to achieve the estimation of the prototypes, the model employs a cross-view prediction task that updates the randomly initialized prototypes. During the inference stage, the incoming data point $\mathbf{x}^{(i)}$ is mapped into feature $\mathbf{h}^{(i)}$ and subsequently compared with its corresponding normal feature $\hat{\mathbf{h}}^{(i)}$ estimated with prototypes. The deviation between $\mathbf{h}^{(i)}$ and $\hat{\mathbf{h}}^{(i)}$ is computed as the anomaly degree score $w^{(i)}$.

to generate local and global views of the crowd flow data. From these views, instances that capture spatial and temporal neighboring information of the original data points are constructed. By maximizing the agreement between the representations of these instances from the two views for each data point, the IC module effectively encodes its comprehensive local and global information. Meanwhile, it enhances the discriminability of the feature space by minimizing the similarity between the representations of different data points.

1) Instance Construction: In our approach, the comparison between positive and negative pairs is vital to the acquisition of the discriminative features. As it is hard to directly compose positive and negative data point pairs without prior knowledge, we employ temporal and spatial augmentation techniques to generate two diverse yet related data views. In the two views, samples from the same data point are defined as positive, and samples from different data points are defined as negative. Subsequently, we construct the spatiotemporal instances carrying both spatial and temporal neighboring information in the two augmented views.

Temporal Augmentation: To enable the model to learn consistent representations from various augmented signals in a contrastive manner, we rely on the approach of simultaneously applying weak and strong augmentation and conducting cross-view tasks. This approach has proven effective in domains such as computer vision [44], [45] and time series [23], [46] representation learning. It is known to promote consistency in representations and enhance the model's generalization capabilities. Specifically, we use weak temporal augmentation to introduce minor variations that do not significantly alter the data's characteristics, and strong temporal augmentation to introduce more pronounced perturbations.

In our method, the weak augmentation method relies on a jitter-and-scale strategy [23], which involves adding Gaussian noise to the time series and applying magnitude changes to each time step using a value drawn from a Gaussian distribution with a mean of 1 and a standard deviation of σ, a hyperparameter. On the other hand, the strong augmentation method uses window warping [42], which has demonstrated effectiveness in time series representation [41]. This strategy involves selecting a random window of the time series and stretching it by a factor of β or contracting it by a factor of $1/\beta$, where β is a parameter. We discuss details about the temporal augmentation selection in Section V-D.4.

Spatial Augmentation: We employ the graph diffusion as the spatial augmentation, which effectively captures the global structure information [26], [27], [47]. By performing graph diffusion, we obtain a global view of the original graph topology. Specifically, we leverage the heat kernel [27], [48] as the instantiation of generalized graph diffusion to power the graph diffusion process. Given an adjacency matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, the global view graph topology can be formulated as:

$$\mathbf{S} = \exp\left(\lambda \odot \left(\mathbf{A}\mathbf{D}^{-1}\right) - \lambda\right) \tag{4}$$

where λ denotes diffusion time; $\odot$ represents element-wise multiplication; $\mathbf{D} \in \mathbb{R}^{n \times n}$ denotes the diagonal degree matrix of $\mathbf{A}$; λ is subtracted element-wise from each element in the resulting matrix.

Instance Sampling: In our method, the strong temporal augmentation creates the local view, while the combination of weak temporal augmentation and graph diffusion generates the global view. These choices are made based on the observed performance of the augmentation techniques in the context of CFAD (detailed in Section V-D.4). For each data point, we sample two sequences of attributed graphs from both local and global views, forming an instance pair. The instance pair is sampled by performing fixed-length sliding window sampling on the temporal dimension, and subgraph sampling on the spatial dimension, and represented as a tuple containing a time series and an adjacency matrix. We employ the $subgraph(\cdot)$ function for spatial sampling. Specifically, given a region v_i, this function identifies the eight nearest regions to v_i in the graph $\mathcal{G}$ and returns nine indices (one for v_i and eight for the neighboring regions):

$$subgraph\left(v_i\right) = \{k \mid v_k \in \underset{\mathcal{V}' \subset \mathcal{V}, |\mathcal{V}'|=9}{\arg\min} \sum_{v_j \in \mathcal{V}} d\left(v_i, v_j\right)\} \quad (5)$$

where $\mathcal{V}'$ represents the set comprising the eight nearest regions to v_i and v_i itself; $d\left(v_i, v_j\right)$ denotes the Euclidean distance between the center point coordinates of region v_i and v_j.

Given a data point $\mathbf{X}_{i,j,:}$, the corresponding spatiotemporal instances y^{local} from the local view and y^{global} from the global view are constructed as:

$$\text{nodes} = subgraph\left(v_j\right) \quad (6)$$

$$y^{\text{local}} = \left(strong\left(\mathsf{X}_{i-T:i,\text{nodes}}\right), \mathbf{A}_{\text{nodes},\text{nodes}}\right) \quad (7)$$

$$y^{\text{global}} = \left(weak\left(\mathsf{X}_{i-T:i,\text{nodes}}\right), \mathbf{S}_{\text{nodes},\text{nodes}}\right) \quad (8)$$

where $weak(\cdot)$ is the weak temporal augmentation function; $strong(\cdot)$ is the strong temporal augmentation function; T is the sequence length.

2) Instance Contrasting: In our approach, we use contrastive learning to derive discriminative representations of spatiotemporal instances. Since each instance consists of a time series and a subgraph adjacency matrix, applying contrastive learning directly is challenging. To address this, we first encode each instance into a vector, which we call a code, that captures both spatial and temporal information. Then we perform contrastive learning on the codes to acquire discriminative representations.

Instance Encoding: As instances are temporal graph signals, we use pre-trained temporal graph encoders that can learn spatial and temporal dependencies effectively [36], [37], [49]. For each view, we train an autoencoder consisting of two GCGRU layers [36] to compress and reconstruct temporal graph signals. The encoder layers of the GCGRU-autoencoder are then utilized to map the instances into codes.

Instance Contrasting: Contrastive learning is then applied to learn the discriminative instance representations. As the two independent GCGRU-encoders encodes instances into different feature spaces, we then apply the feature projection heads to project the codes into an unified feature space, where instance-level contrastive learning is conducted.

In particular, for each view, we construct an MLP with two linear layers and ReLU non-linearity, forming two projection heads. We use $f^{\text{local}}(\cdot)$ and $f^{\text{global}}(\cdot)$ to denote the steps for mapping instances into codes and projecting codes into features within the local and global views, respectively. Codes are mapped into feature $\mathbf{h} \in \mathbb{R}^H$ via $\mathbf{h}^{\text{local}} = f^{\text{local}}\left(y^{\text{local}}\right)$ and $\mathbf{h}^{\text{global}} = f^{\text{global}}\left(y^{\text{global}}\right)$, on which contrastive learning is conducted, where H is the length of feature vector.

Given a batch of N data points $\left\{\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \ldots, \mathbf{x}^{(N)}\right\}$, $2N$ spatiotemporal instances are constructed from its two augmented views. These instances are then encoded and projected into features, resulting in a feature set $\left\{\mathbf{h}^{(1)}, \mathbf{h}^{(2)}, \ldots, \mathbf{h}^{(2N)}\right\}$. Notably, the first N instances correspond to the local view, while the subsequent N instances pertain to the global view. For feature $\mathbf{h}^{(i)}$ $(i \leq N)$, its corresponding feature $\mathbf{h}^{(i+N)}$ from the other view is considered as its positive feature sample, and other features negative. We then employ instance-level contrastive loss to maximize the similarity between the positive feature pairs while minimizing the similarity between the negative feature pairs. In this work, we adopt cosine similarity, i.e.,

$$sim\left(\mathbf{a}, \mathbf{b}\right) = \mathbf{a} \cdot \mathbf{b} / \max\left(\|\mathbf{a}\|_2 \cdot \|\mathbf{b}\|_2, \epsilon\right) \quad (9)$$

where ϵ is a small value to avoid division by zero. This choice is consistent with its common usage in various applications, particularly in the context of contrastive loss computation [21], [22], [23], [40]. We adopt InfoNCE loss [40] to compute instance contrasting loss $\mathcal{L}_{ic}$, i.e.,

$$\mathcal{L}_{ic} = -\sum_{i=1}^{N} \log \frac{\exp\left(sim\left(\mathbf{h}^{(i)}, \mathbf{h}^{(i+N)}\right)/\tau_{ic}\right)}{\sum_{j=1}^{2N} \mathbf{1}_{[j \neq i+N]} \exp\left(sim\left(\mathbf{h}^{(i)}, \mathbf{h}^{(j)}\right)/\tau_{ic}\right)} \quad (10)$$

where $\mathbf{1}_{\text{condition}}$ represents an indicator function, which takes on the value 1 when the condition is true and 0 otherwise. The parameter τ_{ic} controls the temperature, influencing the concentration level of the feature distribution [20].

B. Prototype Contribution Contrast Module

To estimate prototypes coexisting simultaneously in the real-world crowd flow, the PCC module employs a contribution projection head learning the contribution of each prototype to the observed data point. This involves contrasting the contribution vectors between the local and global views. Furthermore, the PCC module employs a cross-view prediction task to iteratively update the randomly initialized prototypes, ultimately leading to an estimation of the prototypes.

Prototype Contribution Contrasting: Contrastive learning in computer vision involves projecting an instance into a feature space with a dimensionality equal to the number of clusters, allowing the feature vector's i-th element to represent the instance's probability of belonging to the i-th cluster [50]. However, the semantic information of crowd flow data used in our work varies relatively mildly, in contrast to images where two distinct pictures (e.g., a dog and a plane) would exhibit significant visual differences. As a result, we do not assign instances to specific clusters, but rather use the i-th element to indicate the relative contribution of the i-th pattern prototype.

To map the features obtained in Section IV-A.2 into contribution vectors, we employ a contribution projection head constructed with two linear layers and ReLU non-linearity, similar to the instance projection head. However, given that the projected vector elements represent the contributions of prototypes, we apply a softmax layer to the MLP. Features are mapped into contribution vectors $\mathbf{z} \in \mathbb{R}^M$ via $\mathbf{z}^{\text{local}} = g\left(\mathbf{h}^{\text{local}}\right)$ and $\mathbf{z}^{\text{global}} = g\left(\mathbf{h}^{\text{global}}\right)$, where M is the number of prototypes, $g(\cdot)$ denotes the contribution projection head.

We concatenate the contribution vectors into matrix form, resulting in $\mathbf{Z}^{\text{local}} \in \mathbb{R}^{N\times M}$ and $\mathbf{Z}^{\text{global}} \in \mathbb{R}^{N\times M}$. Each row of $\mathbf{Z}$ represents the contribution vectors of features, while each column of $\mathbf{Z}$ denotes the contributions of a single prototype through all the features. We conduct prototype contribution contrastive learning on the column vectors. Formally, we denote the prototype contribution vectors (column vectors of $\mathbf{Z}$) with symbol $\mathbf{c}$, forming $\left\{\mathbf{c}^{(1)}, \mathbf{c}^{(2)}, \ldots, \mathbf{c}^{(M)}, \mathbf{c}^{(M+1)}, \ldots, \mathbf{c}^{(2M)}\right\}$ where $\mathbf{c}^{(i)} = \mathbf{Z}^{\text{local}}_{:,i} \in \mathbb{R}^N$ and $\mathbf{c}^{(i+M)} = \mathbf{Z}^{\text{global}}_{:,i} \in \mathbb{R}^N$ $(i \leq M)$ become a positive pair. For instance, $\mathbf{c}^{(1)}$ represents the contributions of the first prototype to all the features in the local view, and $\mathbf{c}^{(1+M)}$ represents the contributions in the global view. The prototype contribution contrasting loss $\mathcal{L}_{pc}$ is computed as Eq. (11).

$$\mathcal{L}_{pc} = -\sum_{i=1}^{M} \log \frac{\exp\left(sim\left(\mathbf{c}^{(i)}, \mathbf{c}^{(i+M)}\right)/\tau_{pc}\right)}{\sum_{j=1}^{2M} \mathbf{1}_{[j\neq i+M]} \exp\left(sim\left(\mathbf{c}^{(i)}, \mathbf{c}^{(j)}\right)/\tau_{pc}\right)} \tag{11}$$

where τ_{pc} denotes the temperature parameter.

Through prototype contribution contrastive learning, we align the prototype contributions for each prototype in the two views while separating the contributions for different prototypes. This process enhances the distinguishable distribution of prototype contribution vectors, thereby indirectly improving the diversity of the learned prototypes.

Pattern Prototype Estimation: We define the prototypes as a learnable parameter $\mathbf{P} \in \mathbb{R}^{M\times H}$, and perform a cross-view feature prediction task to learn it. Specifically, we predict the feature $\mathbf{h}$ with a weighted sum of prototypes, i.e. $\hat{\mathbf{h}} = \mathbf{z}^T\mathbf{P}$. To predict the feature of one view, the contribution vector $\mathbf{z}$ of the other view is adopted as weights, and the prediction loss is defined as Eq. (13).

$$\text{MSE}(\mathbf{a}, \mathbf{b}) = \frac{1}{n}\sum_{i=1}^{n} (\mathbf{a}_i - \mathbf{b}_i)^2 \tag{12}$$

$$\mathcal{L}_{pred} = \frac{1}{N}\sum_{i=1}^{N}\Big(\text{MSE}\left(\mathbf{h}^{(i)}, (\mathbf{z}^{(i+N)})^T\mathbf{P}\right) + \text{MSE}\left(\mathbf{h}^{(i+N)}, (\mathbf{z}^{(i)})^T\mathbf{P}\right)\Big) \tag{13}$$

By training the neural network with $\mathcal{L}_{pred}$, the learnable parameter $\mathbf{P}$ is iteratively updated.

The overall self-supervised loss of our ProtoDetect model is defined as follows.

$$\mathcal{L} = \mathcal{L}_{ic} + \mathcal{L}_{pc} + \mathcal{L}_{pred} \tag{14}$$

C. Anomaly Degree Scoring

In the scoring process, we use symbol Xtest $\in \mathbb{R}^{l_{\text{test}}\times n\times 2}$ to denote the test dataset. For a data point $\mathbf{x} = \mathbf{X}\text{test}_{i,j,:}$, we employ Eq. (7) and Eq. (8) to construct spatiotemporal instances y^{local} and y^{global}. Subsequently, we obtain $\mathbf{h}$ via $\mathbf{h}^{\text{local}} = f^{\text{local}}\left(y^{\text{local}}\right)$ and $\mathbf{h}^{\text{global}} = f^{\text{global}}\left(y^{\text{global}}\right)$, and $\mathbf{z}$ via $\mathbf{z}^{\text{local}} = g\left(\mathbf{h}^{\text{local}}\right)$ and $\mathbf{z}^{\text{global}} = g\left(\mathbf{h}^{\text{global}}\right)$.

We estimate the normal component of $\mathbf{h}$ by a linear combination of prototypes, with the weights indicated by $\mathbf{z}$, i.e. $\mathbf{z}^T\mathbf{P}$. The anomaly score is denoted as $\mathbf{W}_{i,j}$, and we measure the deviation of $\mathbf{h}$ from the estimated normal component using the square distance, as defined in Eq. (15).

$$w = \left(\mathbf{h}^{\text{local}} + \mathbf{h}^{\text{global}} - (\mathbf{z}^{\text{local}})^T\mathbf{P} - (\mathbf{z}^{\text{global}})^T\mathbf{P}\right)^2 \tag{15}$$

We represent the anomaly scores of Xtest with $\mathbf{W} \in \mathbb{R}^{l_{\text{test}}\times n}$.

Aligning with previous works [51], [52], [53], we incorporate a 1-D Gaussian filter to mitigate false positive rates by smoothing anomaly scores, i.e.,

$$G(k) = \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{k^2}{2}\right) \tag{16}$$

$$\widetilde{\mathbf{W}}_{i,:} = \sum_{k=-T/2}^{T/2} G(k) \odot \mathbf{W}_{i-k,:} \tag{17}$$

where $G(k)$ denotes the discrete Gaussian kernel; $\widetilde{\mathbf{W}}$ is the smoothed anomaly scores. During computation, we extend $\mathbf{W}_{i,:}$ by replicating the nearest value on its boundaries.

Utilizing the smoothed anomaly scores, a hyperparameter threshold η is employed to classify points as anomalies (1) or non-anomalies (0). This classification is determined by the condition:

$$O_{i,j} = \begin{cases} 1 & \widetilde{W}_{i,j} \geq \eta \\ 0 & \widetilde{W}_{i,j} < \eta \end{cases} \tag{18}$$

In our subsequent evaluation, the threshold η is set as the K-th highest anomaly score, where the value of K is tailored to align with the evaluation metrics.

V. Experiments

In this section, we evaluate our ProtoDetect method using real-world datasets and compare it with existing works.

A. Datasets and Evaluation Metric

Datasets: We conduct experiments on data from two cities: New York City (NYC) and Chicago. To model the crowd flow dynamics, we utilize taxi trip records and bike trip records, namely NYC Yellow Taxi Trip Records,[1] NYC Citi Bike Trip Records,[2] Chicago Taxi Trip Records[3] and Chicago Divvy Bike Trip Records.[4] By aggregating the trip records every 1 hour, we generate the inflow and outflow features. These crowd flow features are transformed into range [0, 1] by a min-max scaler. To test the performance of CFAD approaches in an online way, we split the dataset into a training set and a testing set.

[1]https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page
[2]https://citibikenyc.com/system-data
[3]https://data.cityofchicago.org/Transportation/Taxi-Trips-2019/h4cq-z3dy
[4]https://data.cityofchicago.org/Transportation/Divvy-Trips/fg6s-gzvg

Preprocessing: Given that the trip records do not encompass the entire city, as observed in previous works [14], [17], [54], our approach centers on a bounding box area that encapsulates both the starting and ending locations of recorded trips. Subsequently, we partition this area into urban regions using a grid system. Specifically, for NYC, we establish 161 regions, each measuring 500 square meters, and for Chicago, we create 87 regions, each with a size of 1000 square meters, as shown in Figure 5.

Following previous works [11], [37], we construct the urban region network by computing the adjacency matrix with a Gaussian kernel. The weighted adjacency matrix $\mathbf{A}$ can be formed as Eq. (19).

$$\mathbf{A}_{ij} = \begin{cases} \exp\left(-\frac{d\left(v_i, v_j\right)^2}{\sigma^2}\right), & i \neq j \text{ and } e_{ij} = 1 \\ 0, & \text{otherwise.} \end{cases} \tag{19}$$

where $d\left(v_i, v_j\right)$ denotes the Euclidean distance between center point coordinates of region v_i and v_j; σ is the standard deviation of distances; e_{ij} indicates the edge between node v_i and v_j, $e_{ij} = 1$ if region v_i and v_j are adjacent.

Ground Truth: Acquiring a comprehensive set of ground truths that records all crowd flow anomalies in a city presents significant challenges. Therefore, in line with previous research [11], [14], [18], [33], [55], [56], we rely on expert-labeled anomaly events to label anomalies for detection. To evaluate the performance of CFAD algorithms in detecting long-term and short-term anomalies, we investigate two sets of events with different durations in NYC and one set of events in Chicago, as depicted in Figure 5 and detailed in Table II. Specifically, the **NYC long-term anomalies (NYC-L)**, previously collected by [55] and utilized as the anomaly ground truth in subsequent studies [11], [14], [18], comprises 20 events with an average duration of 40.4 hours. These events were originally reported by *nycinsiderguide.com* between November 1 and November 30, 2014. In contrast, we adopt the NYC Parks Events[5] as **NYC short-term anomalies (NYC-S)**, which provides event information displayed on the Parks website, *nyc.gov/parks*. The average duration of NYC short-term events is 5.5 hours. Regarding Chicago, we collect Twitter posts posted by *ChicagoAlerts*,[6] an official source that provides notifications on a range of events including sports competitions, festivals, street closures, concerts, parades, and extreme weather. All the posts from August 2019 are gathered to form a relatively objective anomaly set. We refer to this collection as the **Chicago short-term anomalies (Chicago-S)**, encompassing 59 events with an average duration of 5.5 hours. In the anomaly detection stage, if an event persists in region v_j during the time slot t_i, the corresponding data point $\mathbf{x}_{i,j}$ is labeled as anomalous.

Evaluation Metric: Anomaly detection algorithms inherently perform binary classification, distinguishing outliers from inliers. Therefore, we adopt Receiver Operating Characteristic (ROC) and recall as evaluation metrics, in alignment with prior research [11], [18], [51], [56], [57]. To construct the ROC curve, we incrementally vary K from zero to 100% of the total number of data points. The Area Under the Curve (AUC) is then computed to assess the performance of different algorithms. A higher AUC value indicates superior anomaly detection performance. To assess the recall metric, we employ K values at 10%, 20%, 30% and 40% of the total number of data points.

[5]https://data.cityofnewyork.us/City-Government/NYC-Parks-Events-Listing-Event-Listing/fudw-fgrp

[6]https://twitter.com/chicagoalerts

TABLE II
EVENT SAMPLES FROM CROWD GATHERING EVENT DATASETS

	Event Name	Address	Start Time	End Time[1]
NYC-L (20 events, averaging 40.4 hours).				
1	Bowlloween 2014 New York Halloween	624 W 42nd St	31st,21:00	1st,2:00
2	Largest Halloween Singles Party in NYC	247 W 37th St	31st,7:00	1st,3:00
3	Kokun Cashmere Sample and Stock Sale	237 W 37th St	5th,10:30	7th,17:45
4	Big Apple Film Festival	54 Varick St	5th,18:00	9th,23:00
5	InterHarmony Concert: The Soul of Elegiaque	881 7th Ave	6th,20:00	6th,22:00
6	Hiras Master Tailors New York Trunk Show	301 Park Ave	6th,09:00	9th,13:00
7	First Fridays!	881 7th Ave	7th,18:00	7th,22:00
NYC-S (193 events, averaging 5.5 hours).				
1	Union Square Holiday Market	South Plaza	24th,11:00	24th,20:00
2	Exhibition - George Boorujy	Arsenal	13th,09:00	13th,17:00
3	Chair Exercise	J. Hood Wright Recreation Center	8th,09:30	8th,10:30
4	Discovery Walks for Families: North Woods	Charles A. Dana Discovery Center	23rd,11:00	23rd,12:30
5	Grow to Learn Meet Up: Seed Swap!	GrowNYC Green Library	20th,16:00	20th,18:00
6	Health and Race Walking	Fountain Terrace	22nd,09:30	22nd,11:00
7	Sit Aerobics	Reading Room	17th,14:15	17th,15:15
Chicago-S (59 events, averaging 5.5 hours).				
1	Lollapalooza	Grant Park	1st,11:00	1st,23:00
2	Chicago Sky vs New York Liberty	Wintrust Arena	8th,19:00	8th,21:00
3	Retro on Roscoe	Midwest Buddhist Temple	11th,12:00	11th,22:00
4	Thunderstorm	-	18th,6:00	18th,9:30
5	Chicago Air & Water Show	North Beach	18th,13:00	18th,18:00
6	Street Closure	Monroe Harbor	25th,6:00	25th,11:30
7	North Coast Music Festival	Northerly Island	31st,14:00	31st,22:00

[1] NYC-L and NYC-S events occurred in November 2014. Chicago-S events took place in August 2019.

B. Baselines

In our evaluation, we contrast our proposed method against several baseline approaches. These baselines comprise individual methods that exclusively detect anomalies on isolated data point $\mathbf{x}$, as well as spatiotemporal methods that incorporate the spatial and temporal neighbors of $\mathbf{x}$ into the detection process. The details of these baselines are presented below.

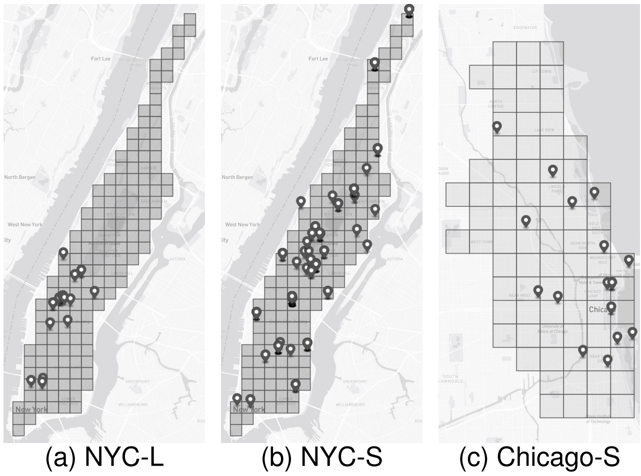


Fig. 5. The urban regions of NYC and Chicago, together with the locations of three sets of anomaly events used in evaluating CFAD algorithms.

Individual methods:

- Elliptic envelope (EE) [28] employs an elliptic envelope to fit the data points and utilizes Mahalanobis distances to quantify the degree of outlyingness.
- Isolation forest (IF) [29] randomly selects features and split values to isolate observations in a tree-like structure. Anomalies tend to have shorter path lengths, and a forest of such random trees that produce shorter paths for specific samples is indicative of anomalies.
- Local outlier factor (LOF) [30] measures the deviation of the density of a given sample in comparison to its neighbors and identifies outliers as samples with substantially lower density than their neighbors.
- One-Class SVM (OCSVM) [31] is an unsupervised learning algorithm that maps the input data into a high-dimensional space and learns a boundary around the normal data points. Any points outside this boundary are considered to be anomalies.

Spatiotemporal methods:

- Spatiotemporal Autoencoder (STAE) is a neural network that compresses signals through a bottleneck structure and subsequently reconstructs the original signal from the compressed features. We utilize STAE with GCGRU [36] layers to represent reconstruction-based anomaly detection methods and capture the spatial and temporal information of the data. The reconstruction error is utilized as the anomaly score.
- STGCN [37] is the first study to utilize purely convolutional structures to extract spatiotemporal features concurrently from graph-structured time series in a traffic prediction study. We employ STGCN as a representative example of prediction-based anomaly detection methods and utilize the prediction error as the anomaly score.
- ST_decomp [18] is a neural network designed specifically for detecting anomalies in urban traffic. It models normal urban dynamics using both spatial and temporal features and subsequently adopts a LOF anomaly detector to detect outliers of abnormal dynamics.
- STGAN [11] is a Generative Adversarial Network (GAN) employed to detect anomalies in traffic speed data and crowd flow data. The generator produces fake graph sequences, while the discriminator distinguishes between them from real sequences. Any sequence that is discriminated as fake is considered anomalous.
- DCdetector, as detailed in [25], stands as a state-of-the-art model for time series anomaly detection. This model employs a multi-scale dual attention contrastive representation learning process, resulting in a permutation-invariant representation with discriminatory capabilities. Given its original design for time series data, we employ one instance of DCdetector for each region in order to conduct spatiotemporal anomaly detection.

C. Implementation Details

In relation to temporal augmentations, we utilize the default parameters for jitter-and-scale and window warping functions as provided by Iwana and Uchida [41].[7] Specifically, our weak temporal augmentation involves the introduction of Gaussian noise with a mean of zero and a standard deviation of 0.03, along with changes in magnitude sourced from a Gaussian distribution with a mean of one and a standard deviation of 0.1. For strong temporal augmentation, we set the value of β to 2. In the context of graph diffusion time (λ), we choose a value of 5, consistent with the approach detailed in [26].

To construct instances, we employ a time series sampling with $T = 12$, ensuring a temporal range that encompasses sufficient neighboring information. The pre-trained spatiotemporal encoders are designed to map instances into tensors with dimensions of 9×16, where each 16-dimensional vector represents an individual region. For feature projection heads, we adopt a Concat-Linear-ReLU-Linear form, featuring two linear layers with 72 and 32 units, respectively, facilitating an effective compression workflow. Similarly, we employ a stack of Linear-ReLU-Linear-Softmax layers for the contribution projection head. The first linear layer consists of 32 units, and the second linear layer accommodates M units.

Based on the findings of our hyperparameter sensitivity analysis as presented in Section V-D.5, we determine the hyperparameters $M = 9$, $\tau_{ic} = 0.1$ and $\tau_{pc} = 0.1$.

D. Results and Discussion

1) Anomaly Detection Performance: The evaluation results are presented in Figure 6 and Figure 7. Our ProtoDetect method outperforms the baseline methods with AUC metrics of 0.85, 0.83 and 0.84 on the three anomaly sets, respectively. Remarkably, the curve of our method almost completely encompasses all other curves, indicating its superior performance across all sensitivity configurations. The STAE method demonstrates suboptimal performance. It is important to note that during both the training and inference processes of STAE, we utilize instances constructed from the original crowd flow data view, employing the same instance construction strategy of ProtoDetect. The suboptimal performance of STAE underscores the advantage of learning spatiotemporal dependencies from instances.

As depicted in Table III, when assessing performance using the recall metric, ProtoDetect consistently attains either the

[7] https://github.com/uchidalab/time_series_augmentation

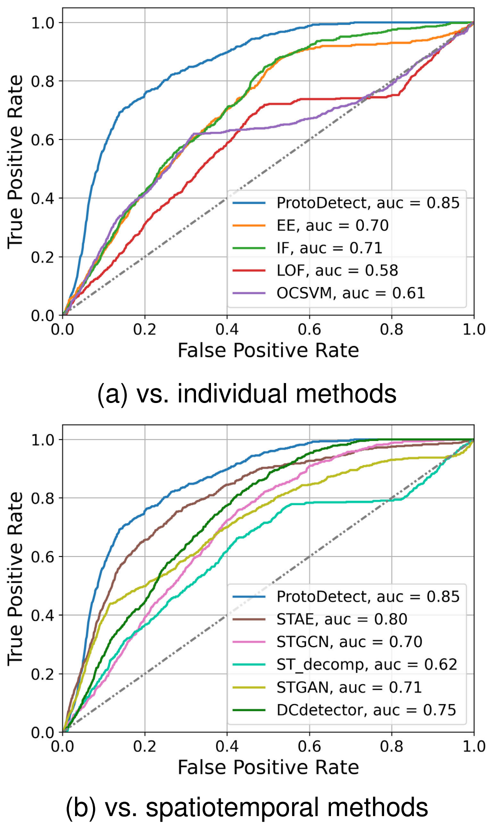


Fig. 6. ROC curves on NYC-L.

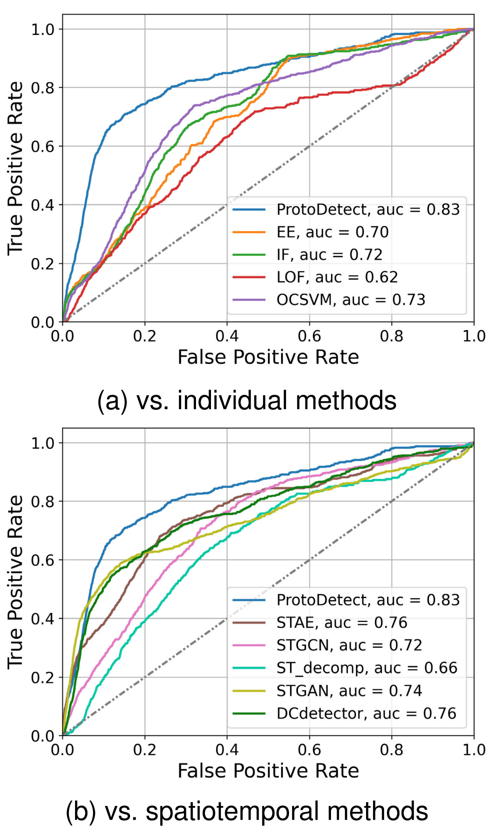


Fig. 7. ROC curves on NYC-S.

highest or the second-highest results across different evaluation settings.

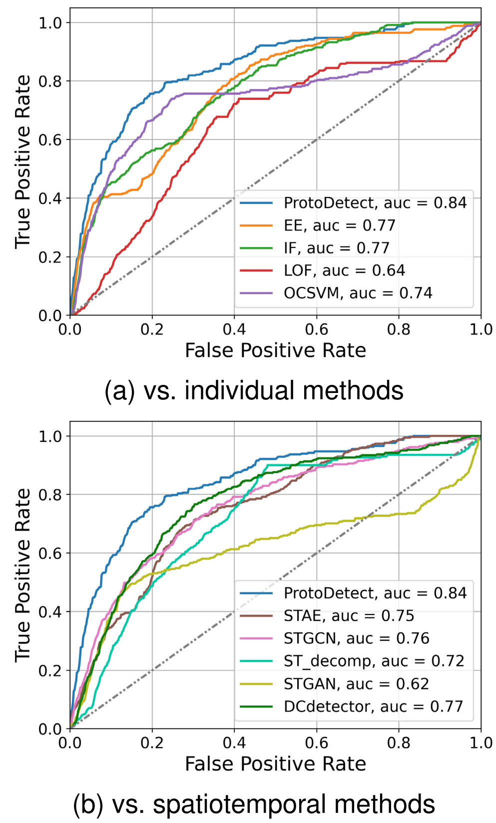


Fig. 8. ROC curves on Chicago-S.

2) Ablation Study: In this section, we aim to examine the individual contributions of our IC and PCC modules to the overall model. To assess the impact of the IC module, we conduct a direct training of the PCC module on the original data using the loss functions $\mathcal{L}_{pc}$ and $\mathcal{L}_{pred}$. We then detect anomalies based on the deviation from prototype-based predictions. Similarly, to evaluate the contribution of the PCC module, we train the IC module solely with $\mathcal{L}_{ic}$. Afterward, we employ general anomaly detection algorithms on the extracted features $\mathbf{h}$.

The results of the ablation study are presented in Table IV. When the IC module is removed, the AUC value decreases by 0.09, 0.10 and 0.09 on the three anomaly sets, respectively. Similarly, when the PCC module is removed, the AUC value decreases by at least 0.05, 0.05 and 0.04 on the three anomaly sets, respectively

It is worth highlighting that the utilization of the IC module leads to improved performance compared to directly employing general anomaly detection algorithms on the original data. As shown in Table V, serving as a front module, IC enhances the AUC metrics of EE, IF, LOF, and OCSVM. This improvement underscores the discriminative nature of the feature space generated by IC, emphasizing its effectiveness in enhancing the detection capabilities of the subsequent anomaly detection algorithms.

3) Effectiveness of Prototype Contribution Contrasting: In this subsection, we investigate the effectiveness of prototype contribution contrasting, i.e. $\mathcal{L}_{pc}$. Specifically, we assess how the inclusion of $\mathcal{L}_{pc}$ impacts CFAD performance in terms of AUC and the diversity of the learned prototypes. To measure prototype diversity, we calculate the standard deviation of

TABLE III
RECALL@K%

Algorithm	NYC-L @10%	NYC-L @20%	NYC-L @30%	NYC-L @40%	NYC-S @10%	NYC-S @20%	NYC-S @30%	NYC-S @40%	Chicago-S @10%	Chicago-S @20%	Chicago-S @30%	Chicago-S @40%
EE	0.21	0.42	0.57	0.70	0.21	0.39	0.57	0.70	0.42	0.46	0.58	0.75
IF	0.22	0.42	0.58	0.70	0.21	0.43	0.66	0.73	0.45	0.56	0.66	0.78
LOF	0.15	0.31	0.44	0.58	0.20	0.37	0.50	0.63	0.15	0.34	0.54	0.71
OCSVM	0.24	0.42	0.58	0.63	0.23	0.51	0.70	0.77	**0.49**	0.67	0.77	0.77
STAE	0.45	0.60	0.76	0.87	0.34	0.56	0.68	0.75	0.36	0.54	0.73	0.79
STGCN	0.17	0.39	0.55	0.72	0.27	0.47	0.63	0.76	0.42	0.59	0.70	0.80
ST_decomp	0.20	0.37	0.49	0.62	0.19	0.39	0.55	0.67	0.25	0.48	0.61	0.74
STGAN	0.39	0.50	0.59	0.70	**0.52**	**0.62**	0.66	0.71	0.39	0.54	0.57	0.62
DCdetector	0.18	0.37	0.53	0.70	0.20	0.41	0.58	0.72	0.39	0.66	0.79	0.85
ProtoDetect	**0.47**	**0.69**	**0.83**	**0.94**	0.44	0.56	**0.73**	**0.81**	0.46	**0.75**	**0.81**	**0.88**

TABLE IV
ABLATION STUDY ON MODEL COMPONENTS

Model	Components	NYC-L	NYC-S	Chicago-S
ProtoDetect	IC + PCC	**0.85**	**0.83**	**0.84**
w/o IC	PCC	0.76	0.73	0.75
w/o PCC	IC + EE	0.80	0.78	0.80
	IC + IF	0.77	0.76	0.78
	IC + LOF	0.62	0.63	0.71
	IC + OCSVM	0.63	0.76	0.76

TABLE V
EFFECT OF THE IC MODULE IMPROVING THE PERFORMANCE OF GENERAL ANOMALY DETECTION ALGORITHMS

Algorithm	NYC-L w/o IC	NYC-L w/ IC	NYC-S w/o IC	NYC-S w/ IC	Chicago-S w/o IC	Chicago-S w/ IC
EE	0.70	**0.80**	0.70	**0.78**	0.77	**0.80**
IF	0.71	**0.77**	0.72	**0.76**	0.77	**0.78**
LOF	0.58	**0.62**	0.62	**0.63**	0.64	**0.71**
OCSVM	0.61	**0.63**	0.73	**0.76**	0.74	**0.76**

pair-wise similarities between prototypes using the similarity matrix $\mathbf{Q}$, as expressed by:

$$\mathbf{Q}_{i,j} = sim\left(\mathbf{P}_i, \mathbf{P}_j\right) \tag{20}$$

$$\text{Diversity} = \sqrt{\frac{1}{M\,(M-1)} \sum_{i=1}^{M} \sum_{j=1,j\neq i}^{M} \left(\mathbf{Q}_{i,j} - \bar{Q}\right)^2} \tag{21}$$

where $\mathbf{Q}$ denotes the similarity matrix of $\mathbf{P}$, and $\bar{Q}$ denotes the mean similarity in the similarity matrix.

Table VI demonstrates that prototype contribution contrasting enhances CFAD performance. Moreover, the prototypes learned with $\mathcal{L}_{pc}$ exhibit greater diversity, aligning with our objective of using prototypes to represent various influencing factors.

4) Temporal Augmentation Selection: Since the appropriate selection of temporal augmentations remains an ongoing challenge, as highlighted in a previous study [58], and the exploration of the combination of temporal and spatial augmentations is relatively uncharted territory, we conduct experiments to investigate these issues in CFAD context.

Our initial investigation centers on the combination of temporal and spatial augmentations within the CFAD framework. As illustrated in Table VII, the optimal performance is observed when the strong temporal augmentation is applied to the local view, and the weak augmentation is applied to the global view. It is important to note that the absence of temporal

TABLE VI
EFFECT OF THE PROTOTYPE CONTRIBUTION CONTRASTING IMPROVING THE DIVERSITY OF ESTIMATED PROTOTYPES

$\mathcal{L}_{pc}$	AUC NYC-L	AUC NYC-S	AUC Chicago-S	Prototype Diversity NYC [1]	Prototype Diversity Chicago-S
w/	0.85	0.83	0.84	0.42	0.53
w/o	0.79	0.79	0.81	0.27	0.40

[1] We train NYC-L and NYC-S on the same dataset, resulting in the same prototype diversity.

TABLE VII
CFAD PERFORMANCES WITH DIFFERENT COMBINATIONS OF SPATIAL AND TEMPORAL AUGMENTATION TECHNIQUES

Temporal Augmentation local	Temporal Augmentation global	AUC NYC-L	AUC NYC-S	AUC Chicago-S
-	-	0.65	0.71	0.61
weak	weak	0.78	0.76	0.70
weak	strong	0.81	0.79	0.75
strong	weak	**0.85**	0.83	**0.84**
strong	strong	0.80	**0.84**	0.80

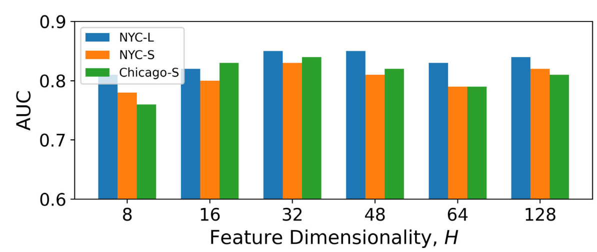


Fig. 9. AUC scores on three anomaly sets, trained with feature dimensions H in {8, 16, 32, 48, 64, 128}.

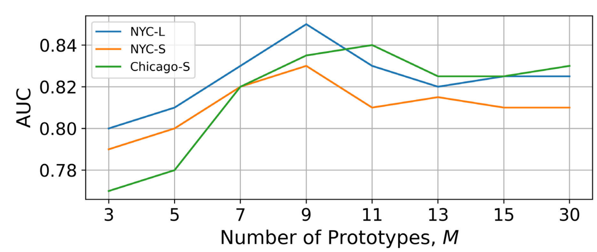
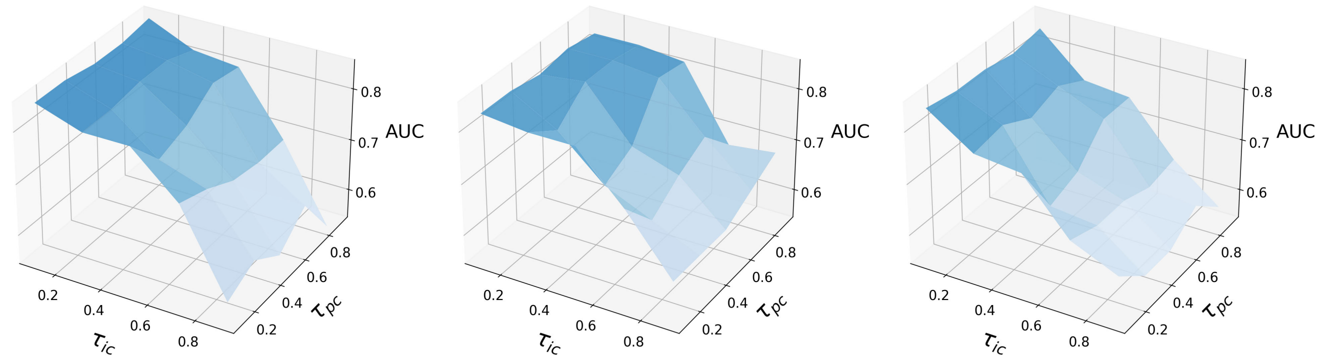


Fig. 10. AUC scores on three anomaly sets, trained with prototype numbers M in {3, 5, 7, 9, 11, 13, 15, 30}.



(a) NYC-L (b) NYC-S (c) Chicago-S

Fig. 11. AUC scores on three anomaly sets, with τ_{ic} and τ_{pc} in {0.1, 0.3, 0.5, 0.7, 0.9}.

TABLE VIII

CFAD PERFORMANCES WITH DIFFERENT STRONG TEMPORAL AUGMENTATION TECHNIQUES

Algorithm[1]	AUC		
	NYC-L	NYC-S	Chicago-S
window warping	**0.85**	**0.83**	0.84
time warping	0.82	0.80	0.79
window slicing	0.83	0.79	**0.86**
permutation	0.75	0.75	0.72
magnitude warping	0.80	0.75	0.78

[1] For each algorithm, we employ the default parameters provided by [41] at https://github.com/uchidalab/time_series_augmentation.

augmentations, as well as the absence of strong augmentations, can result in a significant decline in the AUC metric.

Moreover, we explore the impact of various strong augmentation techniques that introduce significant perturbations. As detailed in Table VIII, our experiments indicate that window warping provides the best performance. Nevertheless, alternative techniques such as time warping, window slicing, and magnitude warping also yield acceptable AUC values. This suggests that while window warping is our preferred choice, the selection of the strong temporal augmentation algorithm is not rigidly fixed. It is worth noting that the permutation algorithm, which shuffles segments of the signal, is not suitable for CFAD, as it severely disrupts the data's fundamental characteristics.

5) Hyperparameter Sensitivity Analysis: We perform sensitivity analysis to study the impacts of several hyperparameters namely, the feature dimensions H, the number of prototypes M, and the two temperature parameters τ_{ic} and τ_{pc}.

Figure 9 illustrates the impact of the feature dimension H on CFAD performance. Results indicate that shorter features offer insufficient information, leading to decreased performance. Optimal performance is achieved when feature lengths are set at 32. Notably, the performance remains stable with increasing feature sizes exceeding 32.

Figure 10 illustrates the relationship between the number of prototypes M and the performance of CFAD. The results show that optimal performance is achieved at $M = 9$ for NYC-L and NYC-S, while for Chicago-S, the optimal performance is observed at $M = 11$. Notably, when M is lower, the performance deteriorates, underscoring the importance of an adequate number of prototypes in capturing the factors that influence normal crowd flows. Additionally, the performance remains stable as M increases beyond the respective optimal values of 9 and 11 for the three sets.

Figure 11 illustrates how the temperature parameters τ_{ic} and τ_{pc} impact the anomaly detection performance. For all the three anomaly sets, as τ_{ic} increases from 0.1 to 0.9, the AUC value decreases significantly, indicating that the feature space becomes less discriminative. However, the varying τ_{pc} does not noticeably affect performance.

6) Case Study: Figure 12 displays individual anomaly score curves for single regions which change over time, particularly during events. When events occur, the background is colored red to signify an anomaly.

We first investigate the anomaly score curves influenced by single events. Figure 12a depicts the anomaly score curve during the Big Apple Film Festival that occurred at 54 Varick St between November 5th, 2014 at 6:00 pm and November 9th, 2014 at 11:00 pm. Figure 12b illustrates the anomaly score curve in Chicago, specifically during the three-day Retro on Roscoe event held at Roscoe Village. This event occurred from 5:00 pm to 10:00 pm on August 9th, 2019, and from 12:00 pm to 10:00 pm on both August 10th and August 11th, 2019. These figures demonstrate a significant increase in anomaly scores during event occurrences. Notably, the anomaly score begins to rise several hours before the events commence, in agreement with the principle of events attracting crowds in anticipation. As the events unfold, the anomaly scores gradually decline to their normal level, suggesting a decrease in the event's attraction.

Additionally, we investigate how a recurring event influences the anomaly score curve. Here we focus on the Exhibition George Boorujy Taxonomy held at the Arsenal Gallary situated at 830 Fifth Avenue, NYC, from 9:00 am to 5:00 pm on October 31st, November 3rd, November 4th, November 5th, and November 6th in 2014. Represented in Figure 12c, the anomaly score curve pertains to the recurring event. As with the other curves, the anomaly score follows a pattern of increasing during event. Notably, a significant decrease in anomaly scores is observed on November 1st and November 2nd, when the periodic event did not occur.

7) Effectiveness of the Prototypes: To demonstrate the effectiveness of the PCC module in our method, and to highlight the coexistence of influencing factors in real-world crowd flows, we present the timeline-averaged prototype

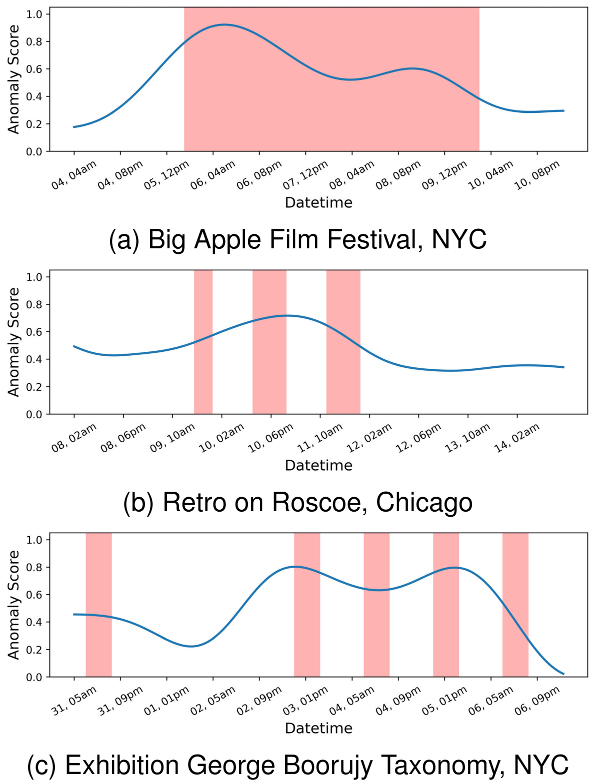


Fig. 12. Single region anomaly scores evolved over time. Light red shaded background represent anomaly events happening in the urban region. The captions of each subfigure reveal the event names.

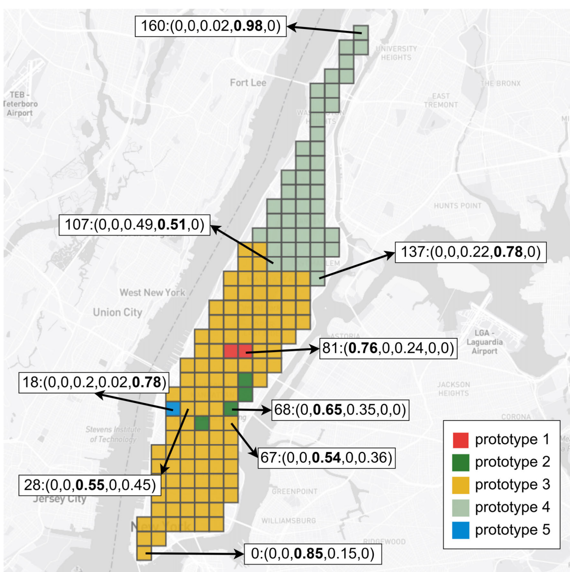


Fig. 13. Average prototype contribution vector of each region. Regions are colored with their major prototypes. Both single (region 0, 160) and mixed (region 28, 67, 68, 107) patterns are observed.

contribution vector of each region of NYC in Figure 13. For clarity in depicting prototype contributions, we set $M = 5$ and use colors to represent the primary contributing prototype of each region. We label the contribution vectors of several representative regions in the format $v : (z_1, z_2, \ldots, z_5)$, where z_i denotes the contribution of the i-th prototype.

The prototype contributions exhibit distinct geographical characteristics, aligning with the reasonable expectation that neighboring regions will share similarities. Notably, the

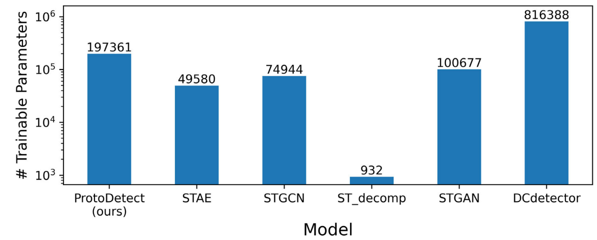


Fig. 14. Number of trainable parameters of deep learning models.

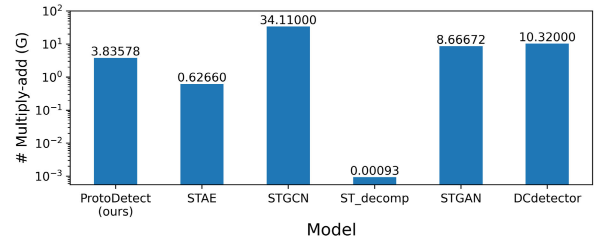


Fig. 15. Number of theoretical multiply-add operations for processing 1000 samples.

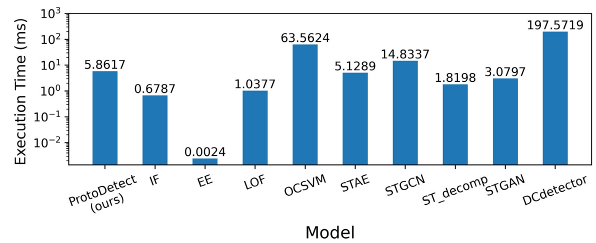


Fig. 16. Empirical runtime performance of baseline models inferring 30-day period anomaly scores of NYC.

top-right regions predominantly correspond to Prototype 4, while the bottom-left regions mostly belong to Prototype 3, creating a noticeable boundary along Central Park North street. Furthermore, we observe that regions farther from these borders generally display a single pattern (e.g., region 0 and region 160), while those closer to the borders exhibit mixed patterns (e.g., region 28, 67, 68, 107), indicating the co-contribution effect from spatial factors. These findings suggest that multiple prototypes simultaneously contribute to the observed crowd flow data.

8) Efficiency Study: In this subsection, we conduct a comparative analysis of our ProtoDetect model's computational efficiency and runtime performance in relation to baseline models.

Figure 14 provides a visual representation of the number of trainable parameters in various deep learning models. Figure 15 offers an illustration of the theoretical count of multiply-add operations necessary for processing a batch of 1000 samples across different deep learning models. Notably, our ProtoDetect model is characterized by 197,361 trainable parameters and demands 3.84 billion multiply-add operations in this specific context. This positions ProtoDetect within a moderate range when compared to the baseline models.

In Figure 16, we provide empirical data illustrating the runtime performance of the baseline models when inferring anomaly scores over a 30-day period for NYC. The experiment is executed on a workstation equipped with an Intel Xeon Gold 5218 CPU and an NVIDIA Tesla V100 GPU. Notably, our proposed method achieves this task within a remarkable

5.86 milliseconds, signifying its potential practical utility in real-world applications.

VI. Conclusion

In this paper, we present ProtoDetect, a novel CFAD framework that learns discriminative crowd flow features and models pattern prototypes in the feature space. These prototypes are subsequently employed to estimate normal crowd flow, providing a baseline for identifying abnormal crowd behaviors. We introduce the IC module to encode spatiotemporal information of crowd flow, improving feature expressiveness and discriminability through contrasting local and global views. Then in the feature space, we propose the PCC module to model prototypes and utilize them to estimate the normal component of crowd flow. Anomaly degree scores are computed by measuring the deviations between incoming crowd flow features and their corresponding normal features. Experimental evaluations on real-world crowd flow anomalies demonstrate the superior performance of ProtoDetect compared to baseline methods, with improved precision in anomaly detection. Ablation studies further validate the effectiveness of both the IC and PCC modules in enhancing the overall performance of the framework. While ProtoDetect exhibits superior performance compared to baseline methods, it currently relies on the pre-training of spatiotemporal encoders, which may introduce practical inconveniences in real-world applications. In our future research endeavors, we aim to enhance ProtoDetect's adaptability for real-world scenarios. This may involve streamlining the workflow, reducing model complexity, and exploring incremental deployment strategies.

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